Threaded Entity Simulation

Project 2

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Introduction

Unlike Project 1, where we use multiple threads to speed up our program, this project uses multiple threads for better programming convenience, then to boost performance. Each entity in our simulation is working on one thread. Our entities in the system are the Watcher: which controls temperature and rainfall, Grain: which controls how much grain is grown based on precipitation; Deer: which controls how much deer is spawned in or removed based off of how much grain or aliens present, and Aliens: which controls how much aliens are spawned in or removed based on deer amount.

The deer quantity has a slight change made to it. When there is an abundance of grain, deer will spawn at double the rate.

Workstation Specifications

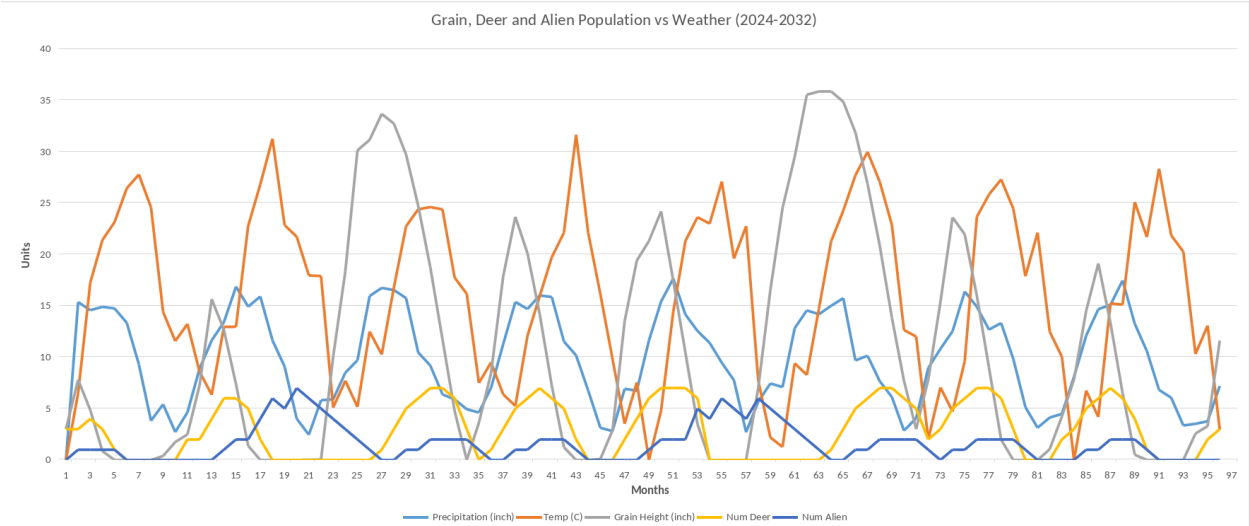
Operating System: Fedora Linux (Kernel 7.2.9) on bare metal
CPU Name: Ryzen 7 9700X
Cores: 8
Threads per core: 2
RAM: 64GB DDR5 (6000 MHz)
Max Clock Speed: 5.6 GHz (Max I can hit before thermal throttling)
Cache:

- L1d: 384 KiB
- L1i: 256 KiB
- L2: 8 MiB
- L3: 32 MiB

Alien Quantity

The quantity I added to the simulation was the number of aliens. Aliens can only spawn if there are at least 3 deer, and rely on eating them. There is also a 10% chance for an alien invasion, which spawns 3 aliens into the simulation. If there's no deer in the simulation, the aliens start to die out. Both deer and aliens have a carrying capacity they move towards each month. Deer and aliens have a two-way interaction where each quantity directly affects the other. More aliens means fewer deer, then less grazing, which allows more grain to grow.

Results



(*Data table at bottom of document*)

In the graph, we have lots of steep rises followed by sharp descents. The grain begins growing in the fall, peaks in winter, and slows down during the spring. No grain grows during the summer months. The trend in the amount of deer is a month behind the trend of the grain. Once grain supply runs low, we see deer begin to fall.

We see something similar with the aliens. The number of aliens follows right behind the amount of deer. But in some months, we see a sharp rise in aliens without a deer population. This discrepancy can be explained by the alien invasion mechanic. We can see these two peaks in months 19 and 52-59, where we have waves of alien invasions, but the alien population falls due to the lack of deer. We can also see that the invasion in month 53 causes the deer to rapidly decrease in a single month. We can see that our aliens drive down the deer population when there is an abundance of grain.

The biggest outliers in the graph are with the large amounts of grain, peaking at months 26 and 61-63. This abundance of grain ties back in with the indirect effect the aliens have on the grain. Right before the grain shot up in these months, we can see waves of alien invasions within the simulation. With these two outliers being explained by our invasion mechanic, we can safely say that the alien entity is affecting both the deer and grain entities.

Conclusion

The simulation is properly doing what we programmed it to do. The precipitation and temperature curves follow the expected seasonal patterns. The grain grows in the cooler, wetter months, and the deer follows the grain trend just a few months behind. The alien population follows the deer population in a similar way, except for when there are invasions of aliens eating the entire deer population. The biggest outliers in the graph can be explained by our invasion mechanic. The alien waves at around 26 and 61 months wiped out the deer population. And with no animals left to graze, the grain grew over 30 inches more than its average yield. Each thread computes its own entity parameters, and with this we can simulate almost anything we could imagine.

Year	Month	Precipitation	Temp C	Grain Height	Num Deer	Num Alien	Temp F
2024	0	0	-17.77777778	3	3	0	0
2024	1	15.375037	6.52529	7.812361	3	1	43.745522
2024	2	14.58576	17.25364889	4.860127	4	1	63.056568
2024	3	14.906654	21.41165389	0.860966	3	1	70.540977
2024	4	14.758124	23.161155	0	1	1	73.690079
2024	5	13.361834	26.43696278	0	0	0	79.586533
2024	6	9.316273	27.79596944	0	0	0	82.032745
2024	7	3.822562	24.60083833	0.000016	0	0	76.281509
2024	8	5.421152	14.40056056	0.392061	0	0	57.921009
2024	9	2.723607	11.57917	1.750245	0	0	52.842506
2024	10	4.630386	13.24242056	2.482722	2	0	55.836357
2024	11	8.897647	8.590956667	7.274301	2	0	47.463722
2025	0	11.650064	6.351405	15.654053	4	0	43.432529
2025	1	13.412707	12.95187	12.67778	6	1	55.313366
2025	2	16.87521	12.97695611	7.384854	6	2	55.358521
2025	3	14.928814	22.78165167	1.385029	5	2	73.006973

2025	4	15.918747	26.83146167	0	2	4	80.296631
2025	5	11.656897	31.28058556	0	0	6	88.305054
2025	6	9.099993	22.86821556	0.000199	0	5	73.162788
2025	7	4.014045	21.73333056	0.000721	0	7	71.119995
2025	8	2.456828	17.96134111	0.018968	0	6	64.330414
2025	9	5.813277	17.90905444	0.047283	0	5	64.236298
2025	10	5.9164	5.106213889	10.061037	0	4	41.191185
2025	11	8.497229	7.714697778	18.357422	0	3	45.886456
2026	0	9.710036	5.17423	30.142216	0	2	41.313614
2026	1	15.965879	12.49698222	31.170677	0	1	54.494568
2026	2	16.749693	10.28304833	33.692116	1	0	50.509487
2026	3	16.547001	16.80964389	32.747269	3	0	62.257359
2026	4	15.762941	22.74679389	29.747437	5	1	72.944229
2026	5	10.468787	24.40291944	24.747467	6	1	75.925255
2026	6	9.193765	24.62539667	18.74749	7	2	76.325714
2026	7	6.386576	24.39718889	11.747517	7	2	75.91494
2026	8	5.900324	17.75976389	4.779981	6	2	63.967575
2026	9	4.941319	16.17719889	0	3	2	61.118958
2026	10	4.620658	7.504268333	3.633867	0	1	45.507683
2026	11	7.037727	9.491136111	8.449932	1	0	49.084045
2027	0	11.280728	6.39061	17.89142	3	0	43.503098
2027	1	15.386649	5.266140556	23.674883	5	1	41.479053
2027	2	14.699982	12.09127222	20.121824	6	1	53.76429
2027	3	16.02133	15.94504278	14.236803	7	2	60.701077
2027	4	15.869856	19.74126611	7.241138	6	2	67.534279
2027	5	11.534691	22.13334389	1.241602	5	2	71.840019

2027	6	10.187101	31.67419	0	2	1	89.013542
2027	7	6.756443	22.11438944	0	0	0	71.805901
2027	8	3.133491	16.14421444	0.088776	0	0	61.059586
2027	9	2.814004	9.839931111	2.876752	0	0	49.711876
2027	10	6.93124	3.537426111	13.510991	2	0	38.367367
2027	11	6.745206	7.546060556	19.414297	4	0	45.582909
2028	0	11.639168	-0.148850556	21.31134	6	1	31.732069
2028	1	15.433832	4.810617222	24.20463	7	2	40.659111
2028	2	17.665176	14.41938611	17.470055	7	2	57.954895
2028	3	14.169477	21.31726556	10.47105	7	2	70.371078
2028	4	12.555569	23.63685611	3.471124	6	5	74.546341
2028	5	11.398597	23.00645167	0	0	4	73.411613
2028	6	9.469409	27.10271222	0.000001	0	6	80.784882
2028	7	7.757334	19.610155	0.006623	0	5	67.298279
2028	8	2.706319	22.76415944	0.006756	0	4	72.975487
2028	9	5.686604	7.724872222	7.036778	0	6	45.90477
2028	10	7.449635	2.189500556	16.573181	0	5	35.941101
2028	11	7.099613	1.271625	24.534706	0	4	34.288925
2029	0	12.831696	9.412231667	29.513187	0	3	48.942017
2029	1	14.56036	8.281981111	35.561584	0	2	46.907566
2029	2	14.194116	14.76187167	35.881416	0	1	58.571369
2029	3	15.012735	21.29014333	35.882366	1	0	70.322258
2029	4	15.753096	24.26753167	34.882393	3	0	75.681557
2029	5	9.694598	27.72039611	31.882393	5	1	81.896713
2029	6	10.148592	29.99071333	26.882393	6	1	85.983284
2029	7	7.675096	27.09049222	20.882393	7	2	80.762886

2029	8	6.138828	22.91316556	13.882557	7	2	73.243698
2029	9	2.860317	12.64932833	7.696377	6	2	54.768791
2029	10	4.065337	12.02046722	3.010369	5	2	53.636841
2029	11	8.965006	2.023536111	7.829226	2	1	35.642365
2030	0	10.820383	7.052156667	15.391743	3	0	44.693882
2030	1	12.539285	4.714679444	23.615818	5	1	40.486423
2030	2	16.400084	9.609305	21.97266	6	1	49.296749
2030	3	14.915723	23.66983222	15.972719	7	2	74.605698
2030	4	12.695763	25.84861333	8.972723	7	2	78.527504
2030	5	13.330187	27.31628444	1.972723	6	2	81.169312
2030	6	9.837823	24.46974	0	3	2	76.045532
2030	7	5.137131	17.89264667	0	0	1	64.206764
2030	8	3.140012	22.13402222	0.000296	0	0	71.84124
2030	9	4.132188	12.49034667	1.04437	0	0	54.482624
2030	10	4.487582	10.05534056	4.23755	2	0	50.099613
2030	11	8.103023	-0.278462222	7.856877	3	0	31.498768
2031	0	12.103476	6.755673889	14.512947	5	1	44.160213
2031	1	14.679064	4.19462	19.133945	6	1	39.550316
2031	2	15.109053	15.24344778	13.345228	7	2	59.438206
2031	3	17.479212	15.12257667	6.515764	6	2	59.220638
2031	4	13.271703	25.11374556	0.515775	4	2	77.204742
2031	5	10.594902	21.70632278	0	1	1	71.071381
2031	6	6.822046	28.35611556	0	0	0	83.041008
2031	7	6.064121	21.87452944	0.000546	0	0	71.374153
2031	8	3.359556	20.28194833	0.002827	0	0	68.507507
2031	9	3.523013	10.32844944	2.572185	0	0	50.591209

2031	10	3.771673	13.08062889	3.298679	2	0	55.545132
2031	11	7.199892	2.95059	11.619843	3	0	37.311062